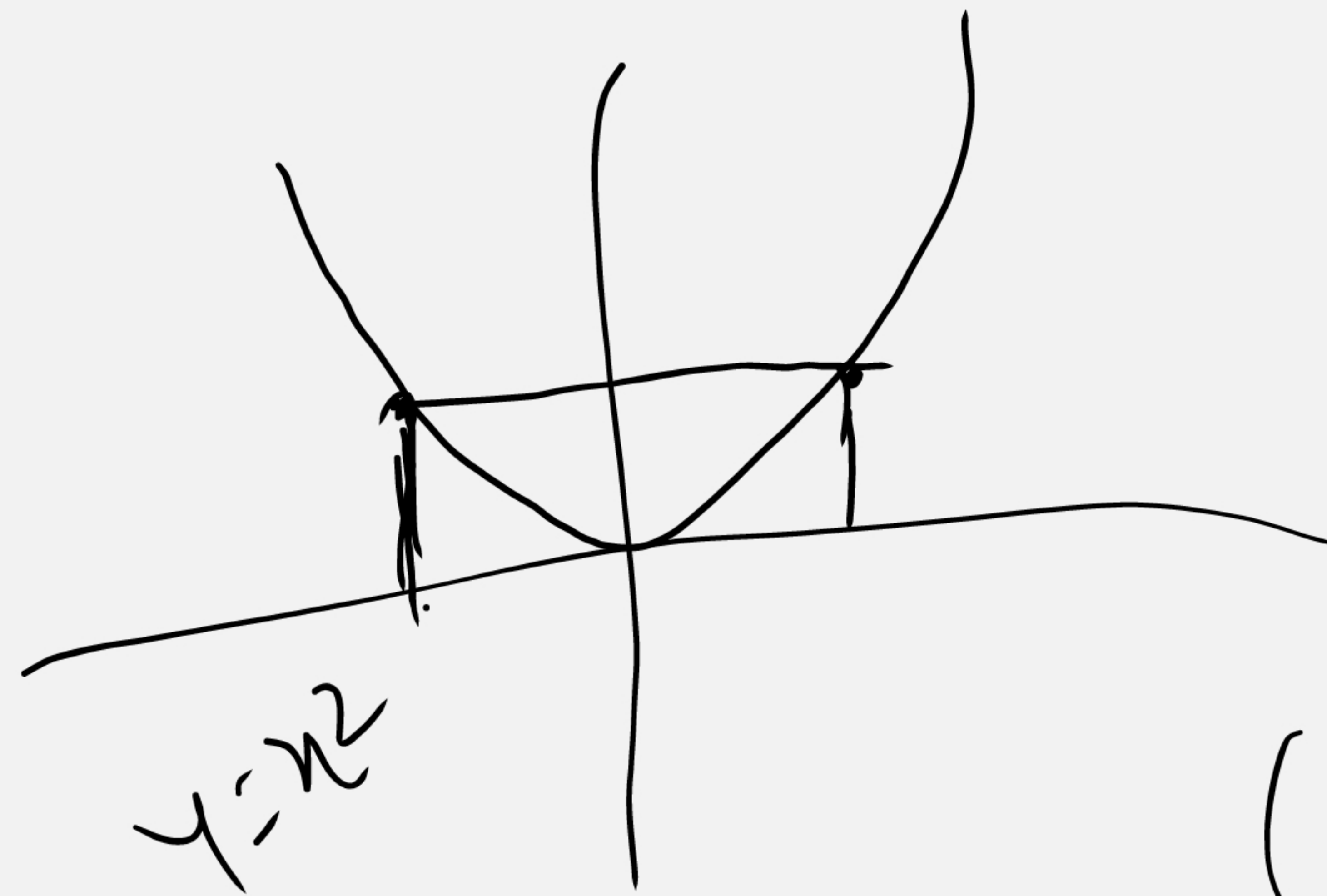




(15 December)

$y: x^2$ $x: 2$

{ domain Range }

Range = domain
onto

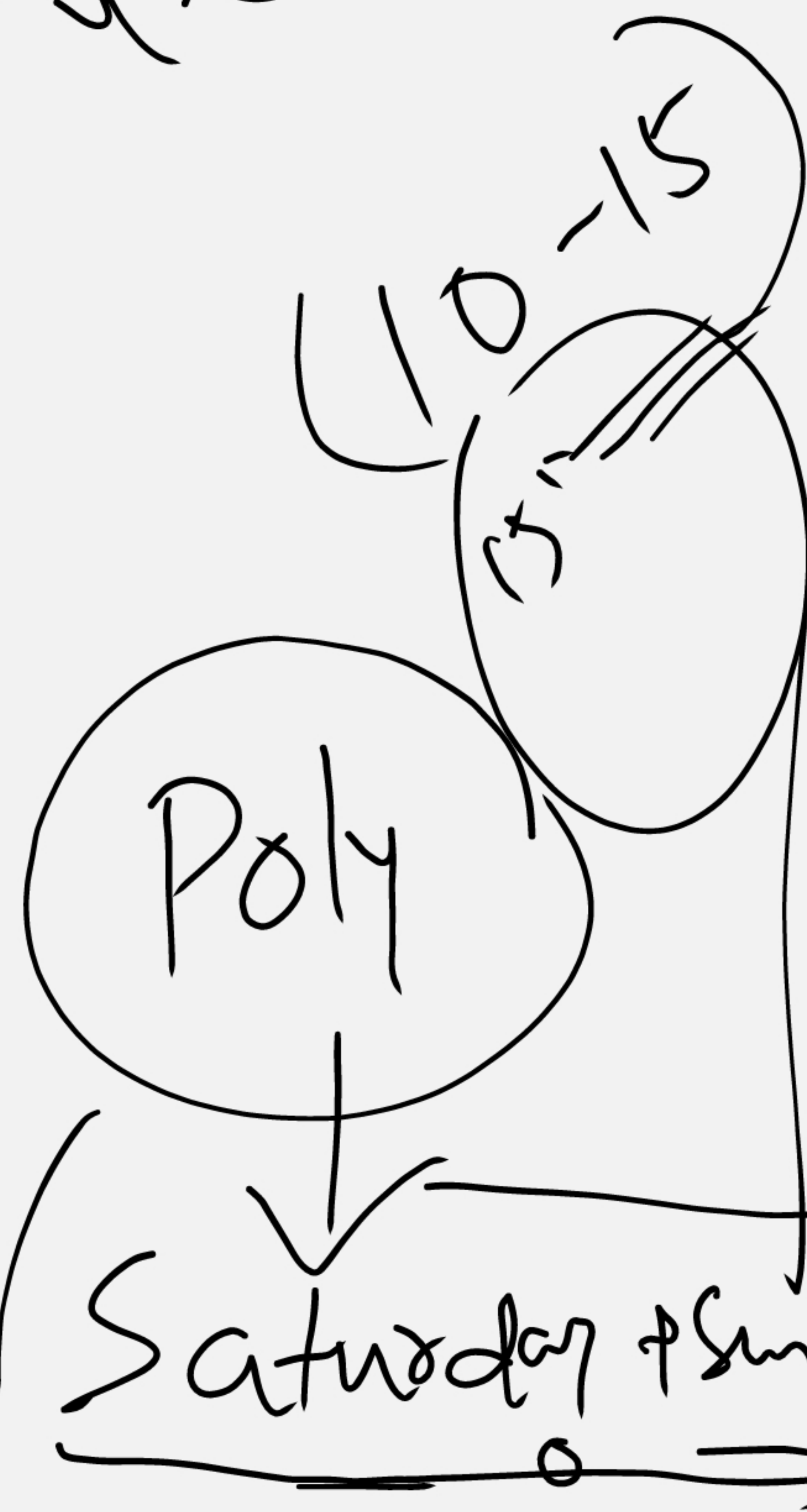
Range + domain onto

1) Domain & Range

2) Maxima-Minima - Range

3) Function KA }
Graphical
Representation

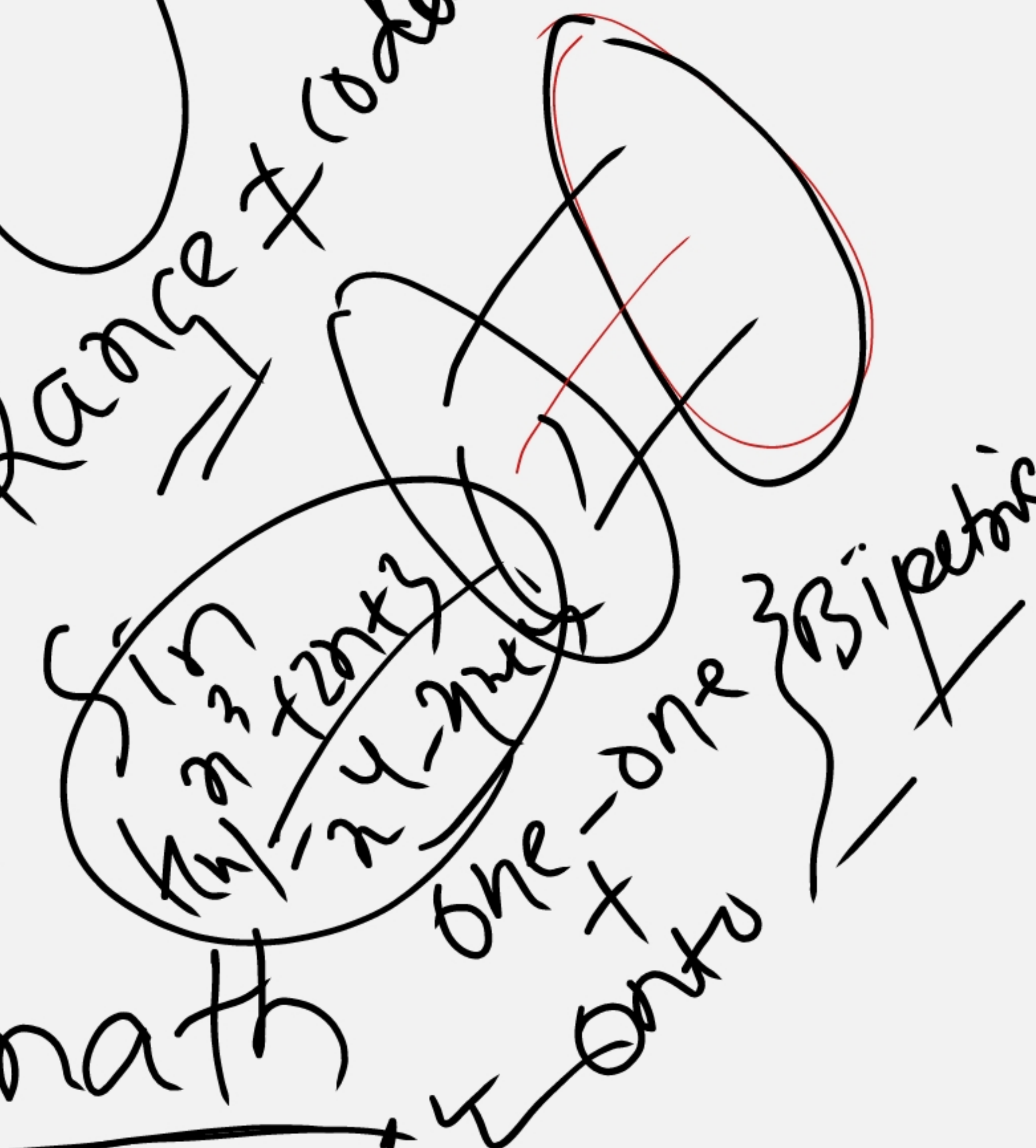
Maxim
Lower -



Function :-

1) Discrete math

{ one-one onto Bijective }



Domain of a function

* Every machine has some limitation in accepting inputs

* The set of all acceptable inputs of machine f is called domain

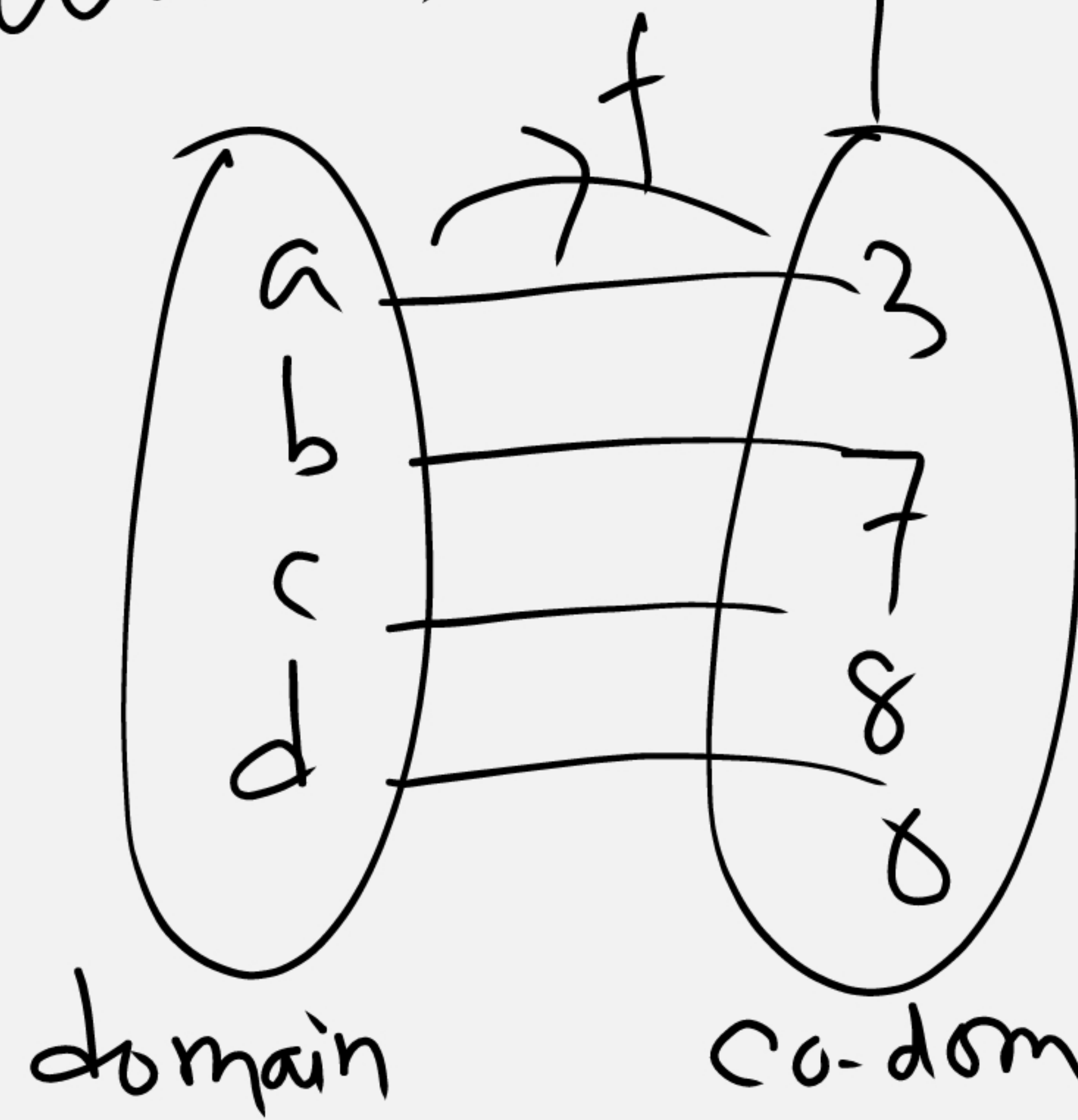
$\{ \}$ → exclude

* The set of all resulting output is called its range.

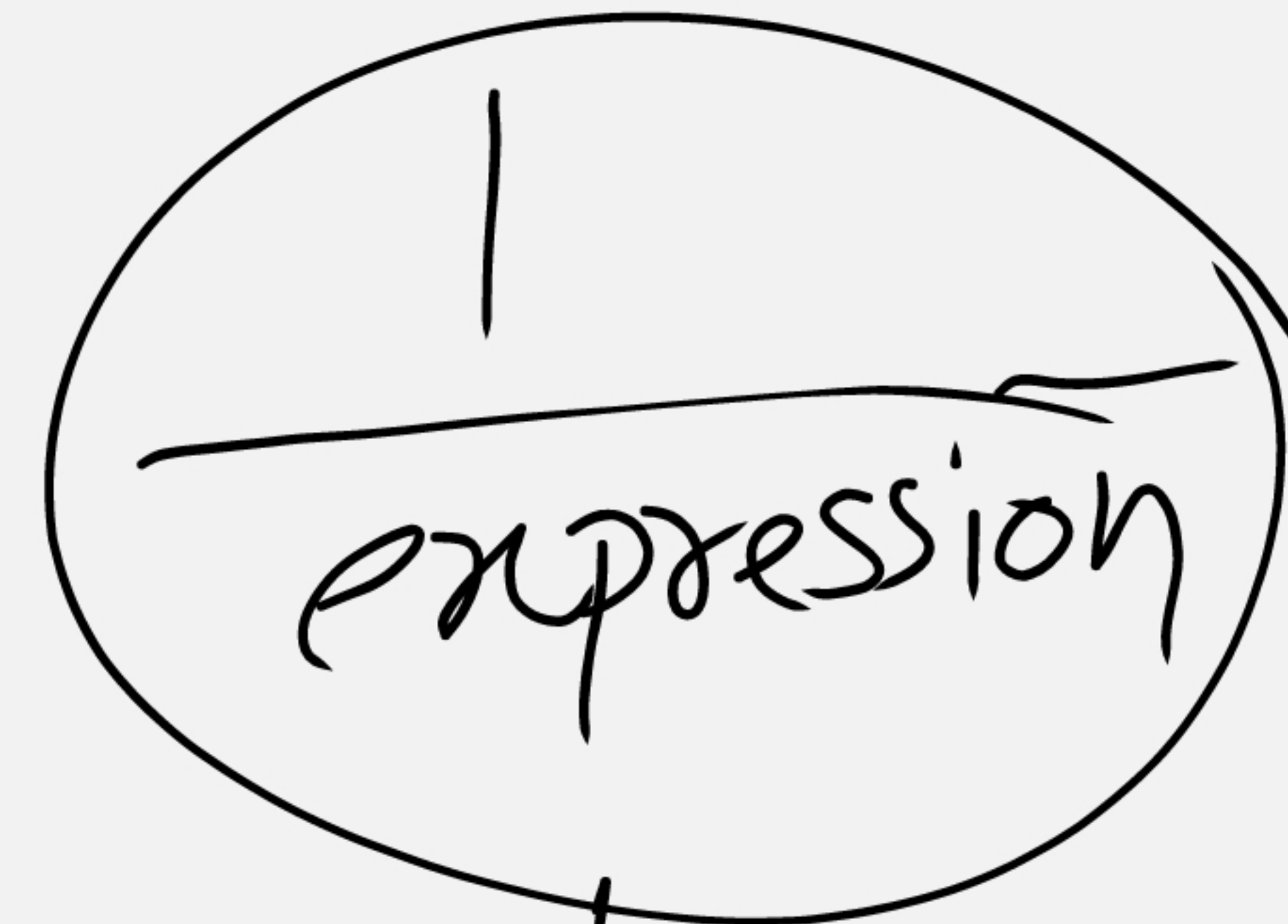
$)$ + expression $\neq 0$

$$x \neq 0$$

$$= \boxed{\mathbb{R} - \{0\}}$$



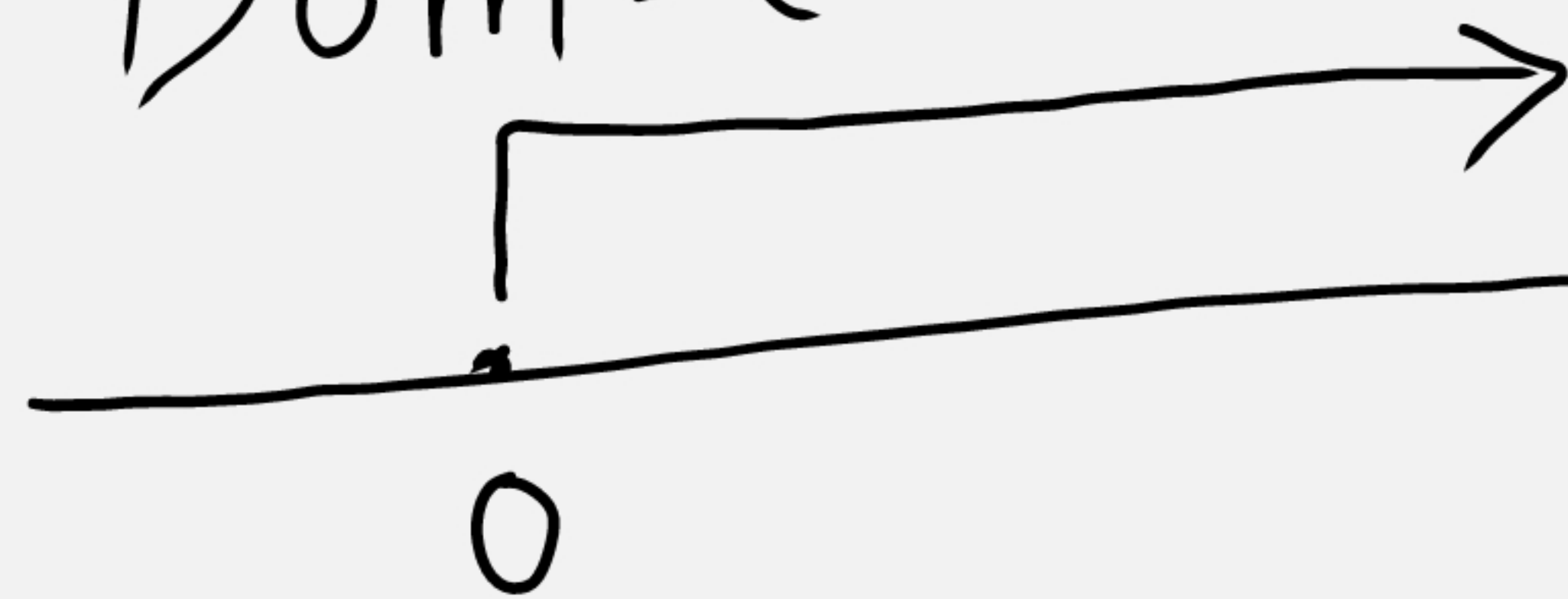
$$\frac{1}{x} \Rightarrow x \neq 0$$



$$\sqrt{\text{expression}} \Rightarrow \underline{\underline{\text{expression} \geq 0}}$$

$$f(x) = \sqrt{x}$$

$$\text{Dom} = [0, \infty)$$



$$y = \sqrt{1-x^2}$$

$$(1-x^2) \geq 0$$

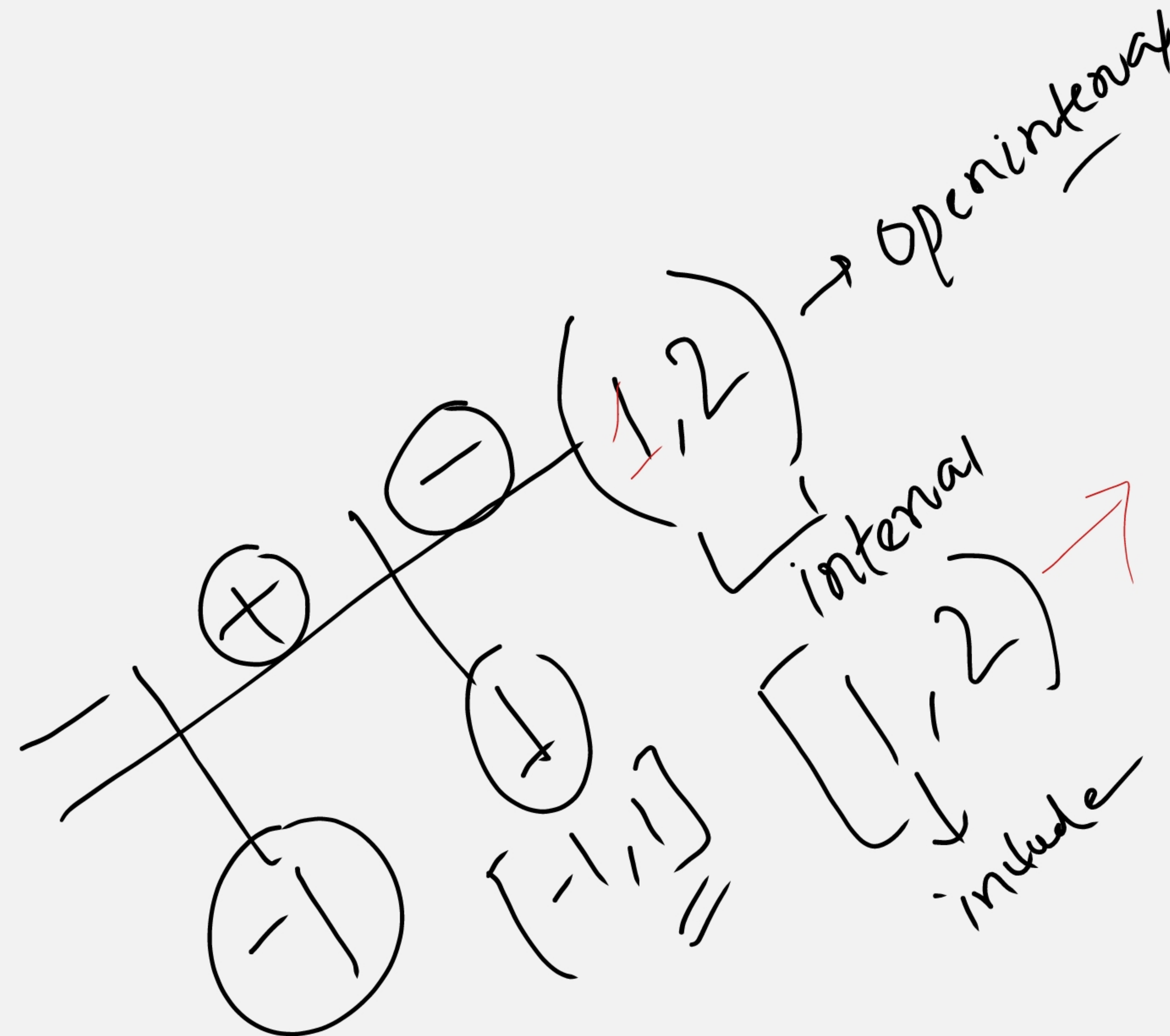
$$(1+x)(1-x) \geq 0$$

$$x = -1, 1$$



Wavy
Curve
method

$$\text{Domain} = [-1, 1] \checkmark$$



$$f_1 = \{ (1, 3), (2, 5), (3, 7) \}$$

$$A = \{1, 2, 3\}$$

$$B = \{3, 5, 7\}$$

$$f: A \rightarrow B$$

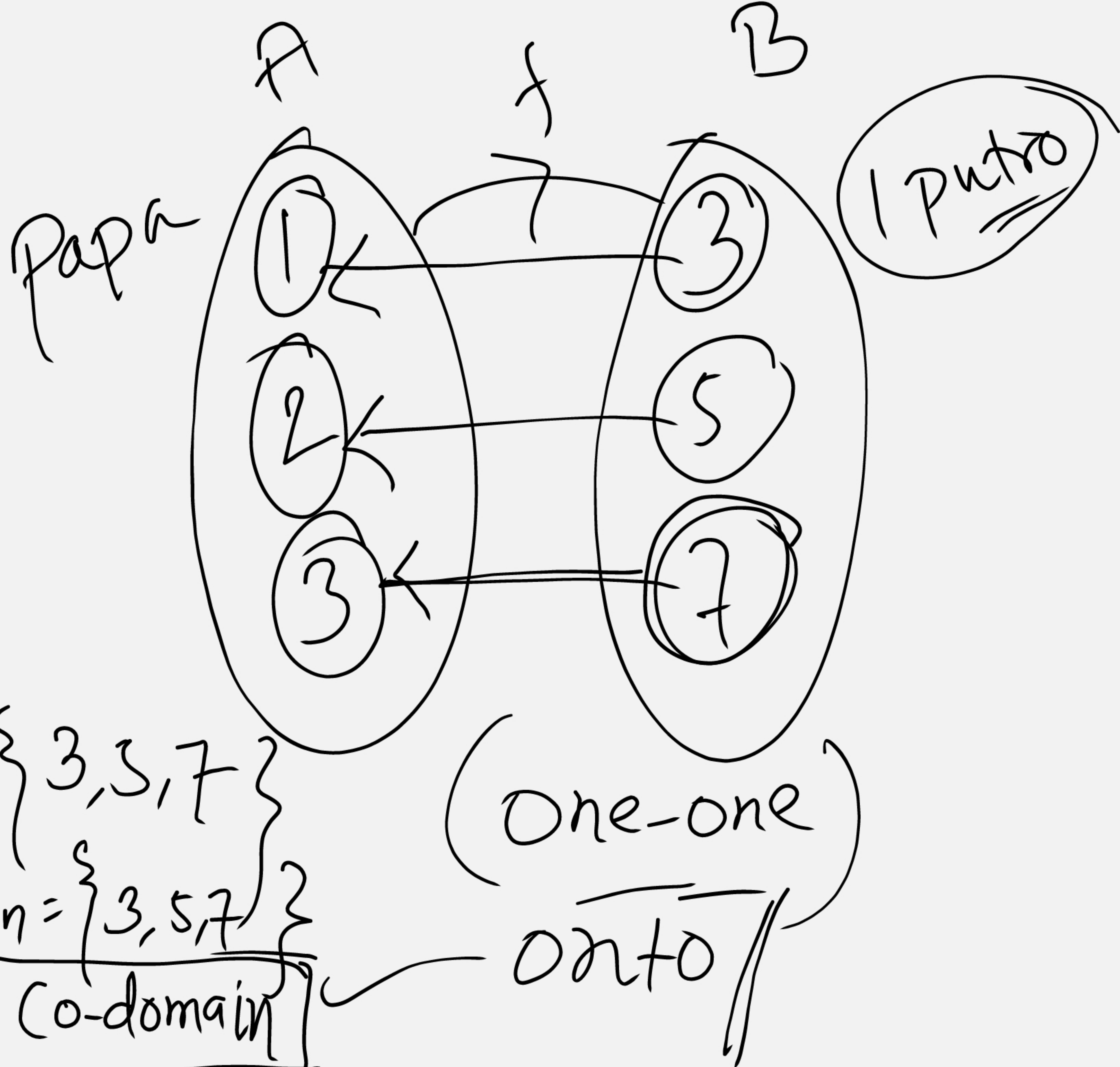
one-one (verified)
 onto

One-one
 onto
 bijective

$$\text{Range} = \{3, 5, 7\}$$

$$\text{Co-domain} = \{3, 5, 7\}$$

$$R = \text{Co-domain}$$



$$f_3 = \{(a, x), (b, x), (c, x), (d, x)\}$$

$$\therefore A = \{a, b, c, d\}$$

$$B = \{x, y, z\}$$

verified?

H/w

$$f: \mathbb{Z} \rightarrow \mathbb{Z} \text{ given by } f(x) = x^2$$